Practice workbook · Tables, graphs and distributions

Constructing frequency tables, bar charts and scatter diagrams and recognising distribution shapes.

9 question parts: 1 worked-example retry and 8 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: PSY-RM-20

1. Question 3 · part c · 3 marks

2025 JUN · Paper 02

Worked-example retry · Attempt part c only.

3 Davina investigated the relationship between the time spent on psychology homework and end of topic test scores.

Davina asked her students to write down the average time (in minutes) they spent on their homework each week and she also recorded their end of topic test scores (out of 30 marks).

Davina's results are shown in **Table 3**.

Participant	Average time spent on homework (in minutes)	End of topic test scores (out of 30 marks)
A	50	17
B	25	10
C	30	15
D	20	10
E	55	25
F	15	9
G	80	30
H	50	19
I	60	27
J	20	14

Table 3

(a) Calculate the range for the **end of topic test scores**.

(1)

Range .....

(b) Calculate how many students spent **more than** 30 minutes on their homework as a fraction of all students.

You must give your answer in the lowest form.

(2)

Fraction .....

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1. Question 3 · part c · 3 marks

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Worked-example retry · Attempt part c only.

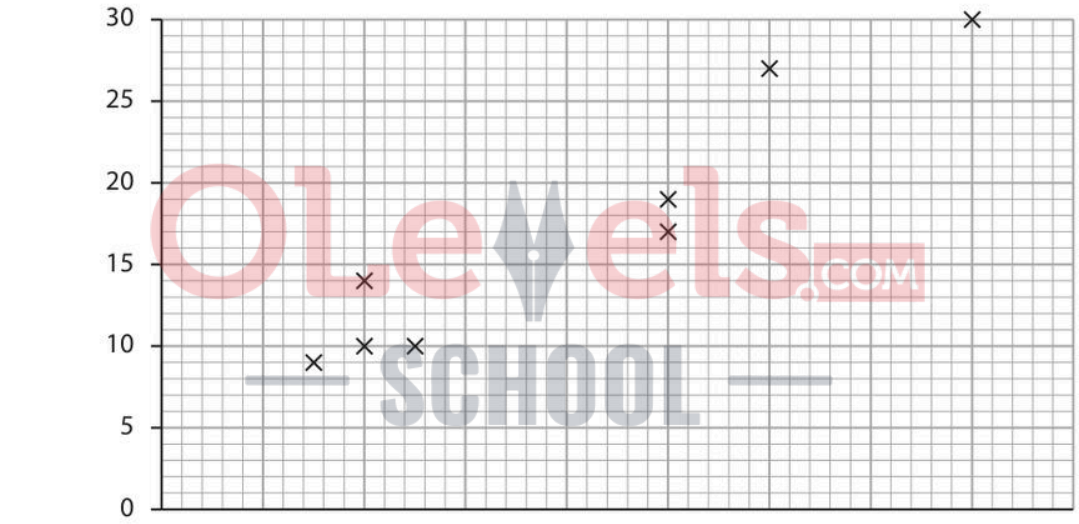
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(c) Complete the scatter diagram for participants C and E using the data in **Table 3**, including the label and scale for the x-axis.

(3)

A scatter diagram to show the relationship between average time spent on homework and end of topic test score



(Total for Question 3 = 6 marks)

1. Question 3 · part c · 3 marks · Working and answer

Working and answer

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2. Question 1 · part d.ii · 3 marks

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Further practice · Attempt part d.ii only.

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1 Matthew investigated the effectiveness of drug treatment for patients with depression. He interviewed six patients from a local mental health service provider before and after drug treatment.

Matthew only used preset questions in his interview.

(a) Identify the type of interview Matthew used in his investigation.

(1)

☒ **A** Structured interview

☒ **B** Unstructured interview

☒ **C** Semi-structured interview

☒ **D** Covert interview

(b) Matthew made sure that the participants in his interviews had the right to withdraw.

Define what is meant by the 'right to withdraw'.

(1)

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(c) Give **two** ways that Matthew could ensure the confidentiality of the patients he interviewed.

(2)

1 .....

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2 .....

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2. Question 1 · part d.ii · 3 marks

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Further practice · Attempt part d.ii only.

(d) Participant responses to the question 'How often do you feel tired?' were scored using a scale of 0 to 5, with 0 being 'never feel tired' and 5 being 'always feel tired'.

The results of Matthew's investigation are shown in **Table 1**.

Participant	Score (out of 5) for tiredness before drug treatment	Score (out of 5) for tiredness after drug treatment
A	5	3
B	4	3
C	5	2
D	3	0
E	5	1
F	4	1

Table 1

(i) The score for tiredness of **Participant F** decreased from before drug treatment to after drug treatment.

Calculate the percentage decrease in the score for tiredness of **Participant F**.

(2)

Percentage decrease in score for tiredness of **Participant F** .....

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2. Question 1 · part d.ii · 3 marks

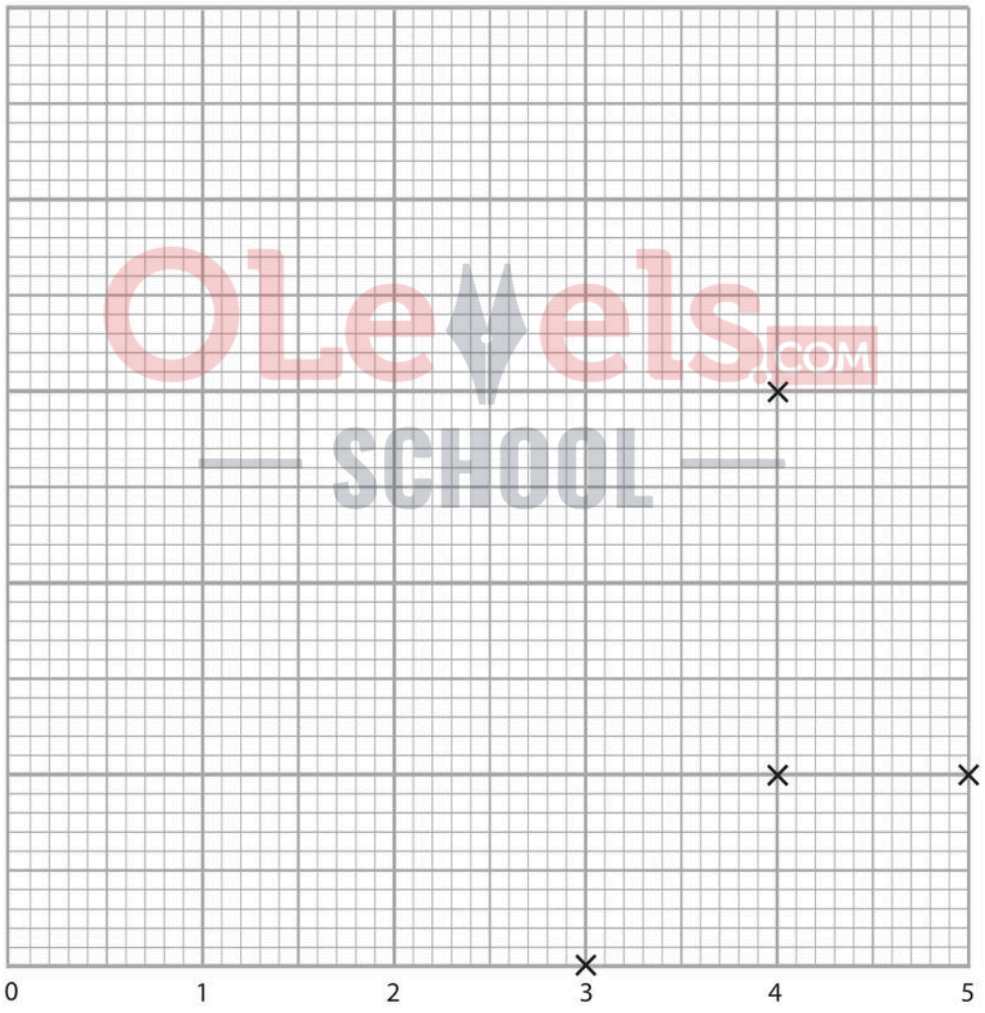
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Further practice · Attempt part d.ii only.

(ii) Complete the scatter diagram for **Participants A and C**, using the data in **Table 1**, including the label and scale for the y-axis.

(3)

A scatter diagram to show the relationship between tiredness and drug treatment



Score (out of 5) for tiredness before drug treatment

(Total for Question 1 = 9 marks)

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2. Question 1 · part d.ii · 3 marks · Working and answer

Working and answer

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3. Question 4 · part whole · 1 mark

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Further practice · Attempt part whole only.

4 **Figure 1** shows a distribution of data presented in a histogram.

A histogram to show the frequency distribution of scores for self-reported time spent (in minutes) per day on social media by 14 year old students

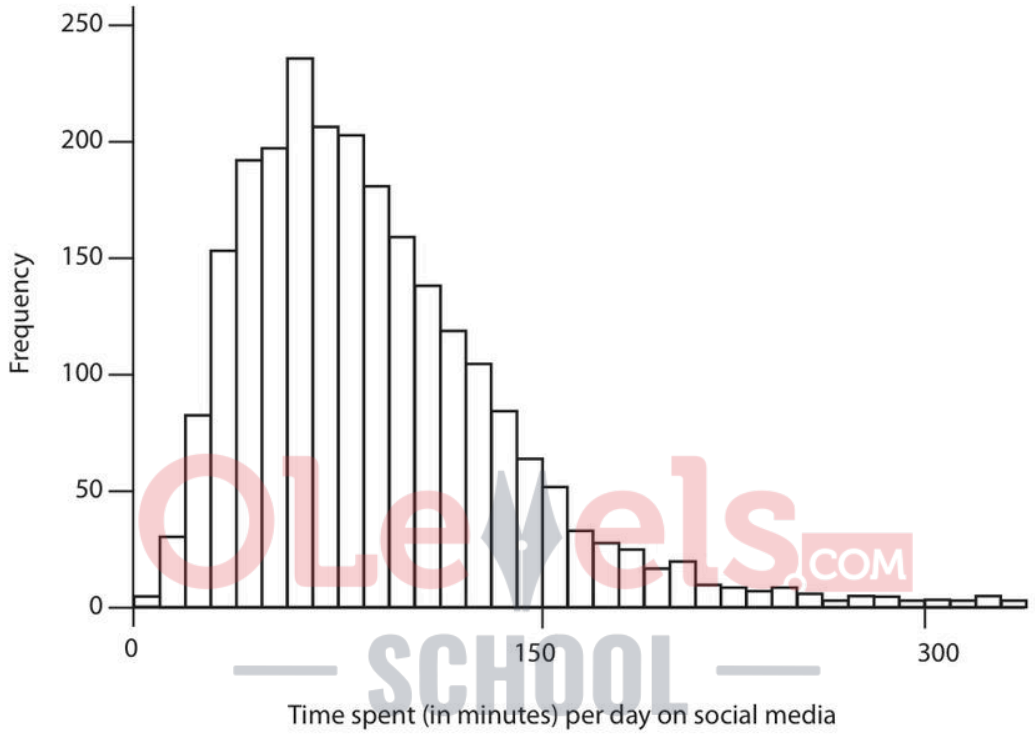


Figure 1

Name the type of skew shown in **Figure 1**.

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(Total for Question 4 = 1 mark)

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Working and answer

4. Question 4 · part b · 3 marks

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Further practice · Attempt part b only.

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4 Ricky used a field experiment to investigate whether there is a difference between English and Sociology students' obedience to an authority figure.

He arranged for an English teacher and Sociology teacher to give their students an instruction to complete a piece of work. The teachers then left the classroom for 45 minutes.

Ricky recorded the number of students who did and did not complete the work. The results are shown in **Table 3**.

	Number of students who did complete the work	Number of students who did not complete the work
English students	16	8
Sociology students	15	10

Table 3

(a) Calculate the percentage of students who completed the work, out of all the students in the experiment.

(2)

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Percentage of students who completed the work .....

4. Question 4 · part b · 3 marks

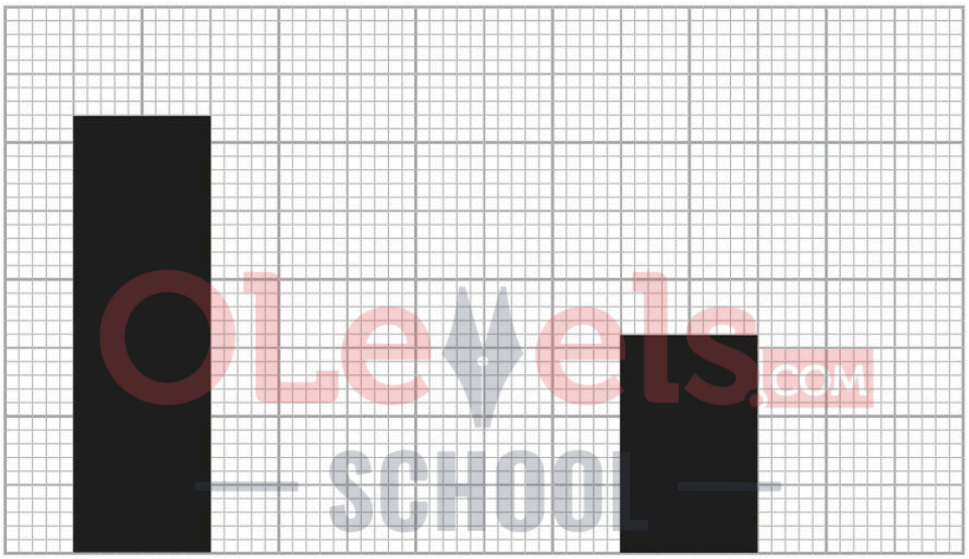
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Further practice · Attempt part b only.

(b) Complete the bar chart with the data from **Table 3** for the Sociology students, including the label and scale for the y-axis.

(3)

A bar chart to show the number of English and Sociology students who did and did not complete the work set by the teacher.



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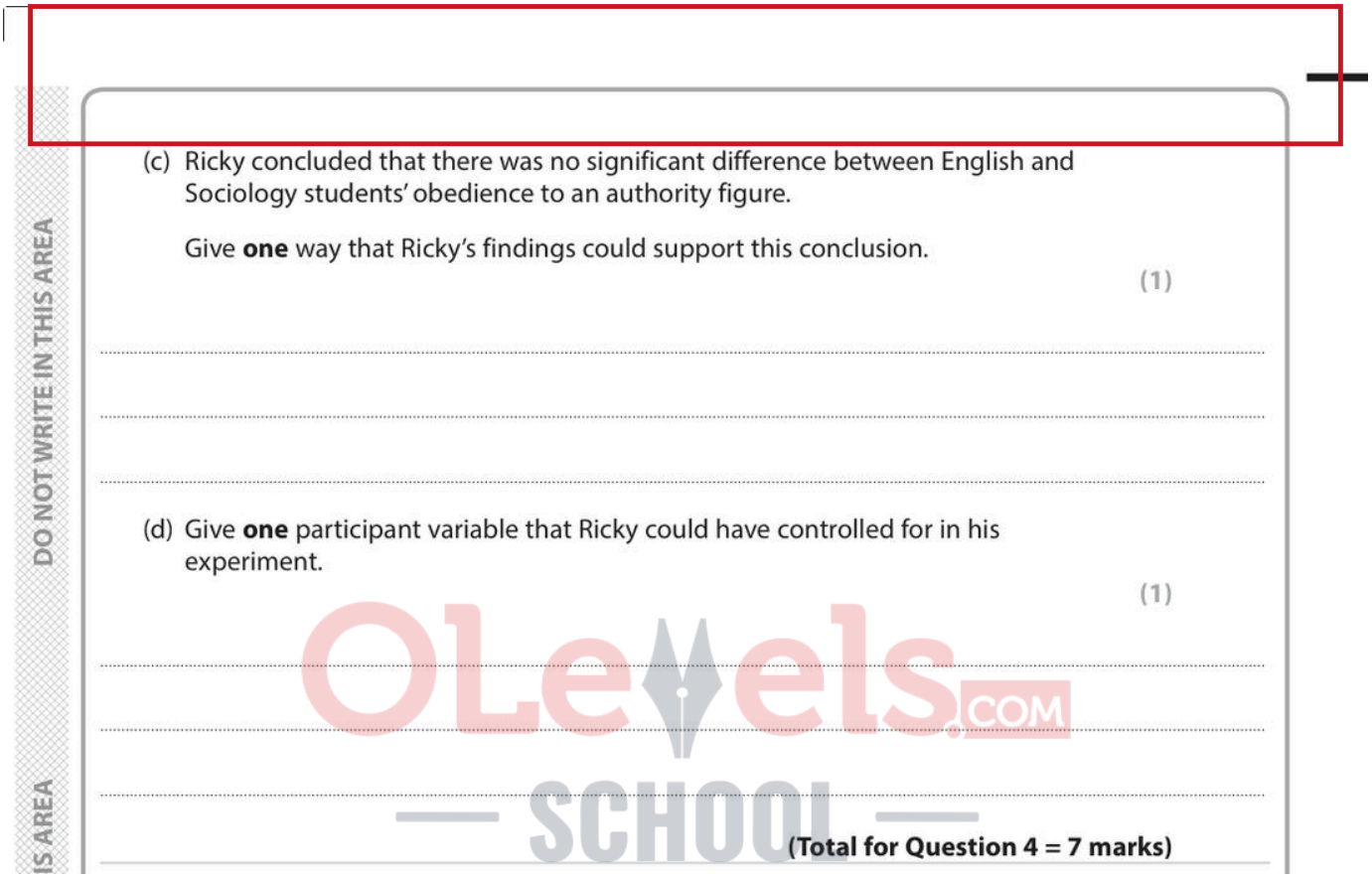
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4. Question 4 · part b · 3 marks

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Further practice · Attempt part b only.



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(c) Ricky concluded that there was no significant difference between English and Sociology students' obedience to an authority figure.

Give **one** way that Ricky's findings could support this conclusion.

(1)

(d) Give **one** participant variable that Ricky could have controlled for in his experiment.

(1)

(Total for Question 4 = 7 marks)

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Working and answer

5. Question 5 · part c · 3 marks

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Further practice · Attempt part c only.

5 Cathy researched whether short-term memory improved when grouping was used for learning telephone numbers. She showed the participants 10 telephone numbers.

- Condition A learned whole telephone numbers, for example 01234567890
- Condition B learned grouped telephone numbers, for example 0123 4567 890

Each telephone number was shown for 15 seconds. Then, after a 30-second delay, participants were asked to write the telephone number down.

The results of Cathy's experiment are shown in **Table 3**.

	Number of correctly recalled telephone numbers	Number of incorrectly recalled telephone numbers
Condition A	3	7
Condition B	6	4

Table 3

(a) Identify the independent variable for Cathy's experiment.

(1)

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(b) Cathy concluded that grouping improves recall in short-term memory.

Give **one** way that the data in **Table 3** could support this conclusion.

(1)

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5. Question 5 · part c · 3 marks

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Further practice · Attempt part c only.

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(c) Cathy analysed the results in more detail to see if the age of the participants had any impact on correct recall.

The results of her analysis are shown in **Table 4**.

	Number of correctly recalled telephone numbers	Number of incorrectly recalled telephone numbers
Condition A Aged 18–29	0	5
Condition A Aged 30–49	3	2
Condition B Aged 18–29	3	3
Condition B Aged 30–49	3	1

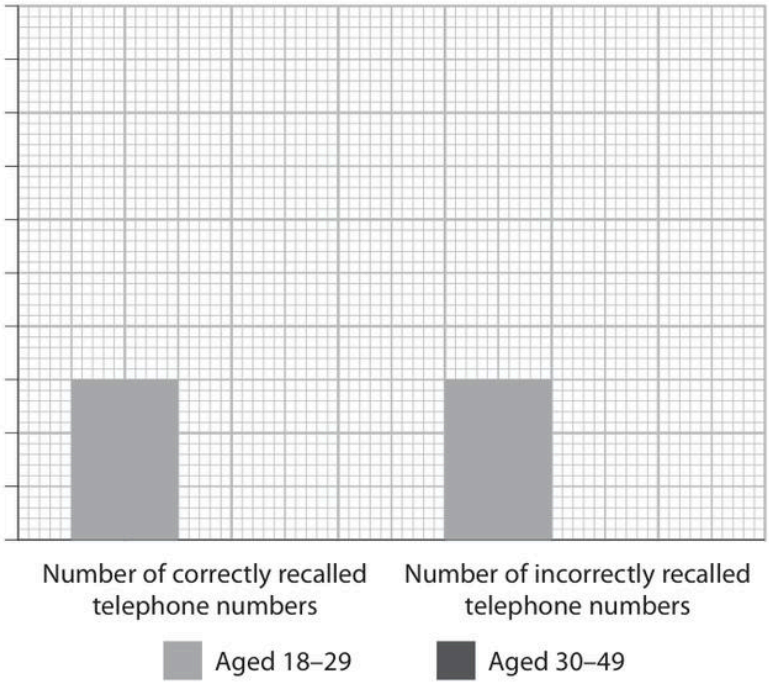
Table 4

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Complete the bar chart with the data for Condition B from **Table 4**, including the label and scale for the y-axis.

(3)

A bar chart to show the number of correctly and incorrectly recalled telephone numbers in Condition B by age



(Total for Question 5 = 5 marks)

5. Question 5 · part c · 3 marks · Working and answer

Working and answer

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6. Question 2 · part b · 3 marks

2022 JUN · Paper 02

Further practice · Attempt part b only.

2 Toby is investigating the impact of damage to the left hemisphere of the brain.

He asked patients with damage to their left hemisphere to complete a questionnaire about the impact it has had on them.

His results are shown in **Table 2**.

Patient	Language difficulty	Limited mobility in the right arm	Limited mobility in the right hand	Difficulties with the right visual field
A	Yes	No	No	Yes
B	Yes	Yes	No	Yes
C	Yes	Yes	No	No
D	Yes	Yes	Yes	Yes
E	Yes	Yes	Yes	No

Table 2

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(a) Calculate how many patients had difficulties with the right visual field as a fraction of all patients.

(1)

Fraction .....

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6. Question 2 · part b · 3 marks

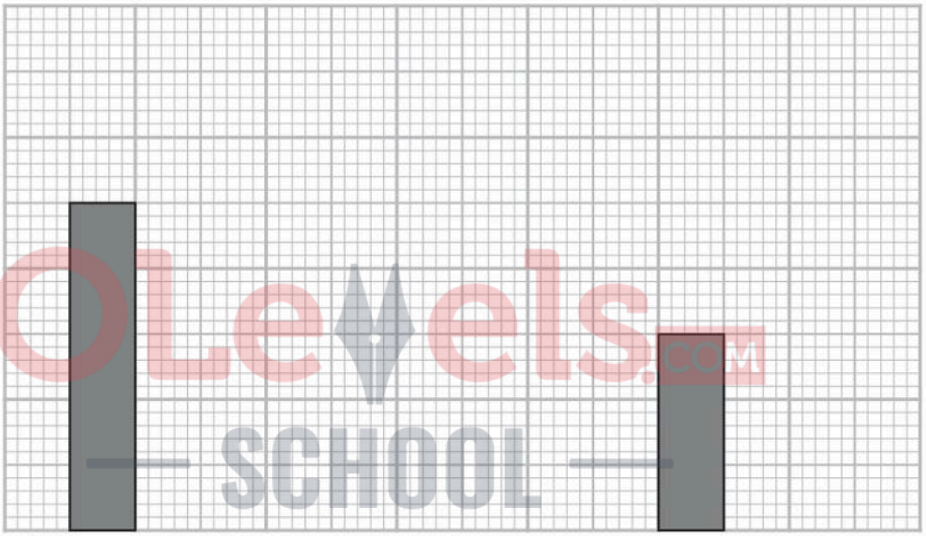
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Further practice · Attempt part b only.

(b) Complete the bar chart using the data in **Table 2**, including the label and scale for the y-axis.

(3)

A bar chart to show the number of patients reporting impacts from damage to the left hemisphere of the brain



Impact of damage to the left hemisphere

(Total for Question 2 = 4 marks)

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6. Question 2 · part b · 3 marks · Working and answer

Working and answer

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7. Question 2 · part b · 3 marks

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Further practice · Attempt part b only.

2 Kamira is investigating morality. She wants to find out how likely people are to return lost items. Kamira conducted an observation for her investigation.

Kamira went to a large shopping centre and positioned herself in the food court where people often sit to eat and have been known to leave personal items behind when leaving.

When someone had left an item behind, Kamira observed the behaviour of all passers-by who saw the item. She recorded whether they:

- looked at the item, but then ignored it
- picked up the item and gave it to a member of staff
- walked away with the item.

(a) Describe why Kamira may have used an observation research method for her investigation.

(2)

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7. Question 2 · part b · 3 marks

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Further practice · Attempt part b only.

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(b) Draw a frequency table that Kamira could have used to record the data for her observation.

You must include appropriate column and row headings.

(3)

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(c) Describe **one** situational variable that Kamira may have considered when conducting her observation.

(2)

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(Total for Question 2 = 7 marks)

7. Question 2 · part b · 3 marks · Working and answer

Working and answer

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8. Question 4 · part c · 3 marks

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Further practice · Attempt part c only.

4 Pedro was investigating whether schemas have an influence on memory accuracy. He decided to test four students from his psychology class.

He placed 10 typical objects, such as a chair, and 10 unusual objects, such as a laundry basket, in a classroom. The students had to enter the room and look around for one minute.

Pedro recorded the number of typical objects and unusual objects students could remember.

The results of Pedro's investigation are shown in **Table 1**.

	Number of typical objects recalled (out of 10)	Number of unusual objects recalled (out of 10)
Student A	8	3
Student B	10	5
Student C	9	5
Student D	5	2

Table 1

(a) Calculate the percentage of students who recalled **more than five typical objects**, out of all the students in the investigation.

(1)

Percentage .....

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8. Question 4 · part c · 3 marks

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Further practice · Attempt part c only.

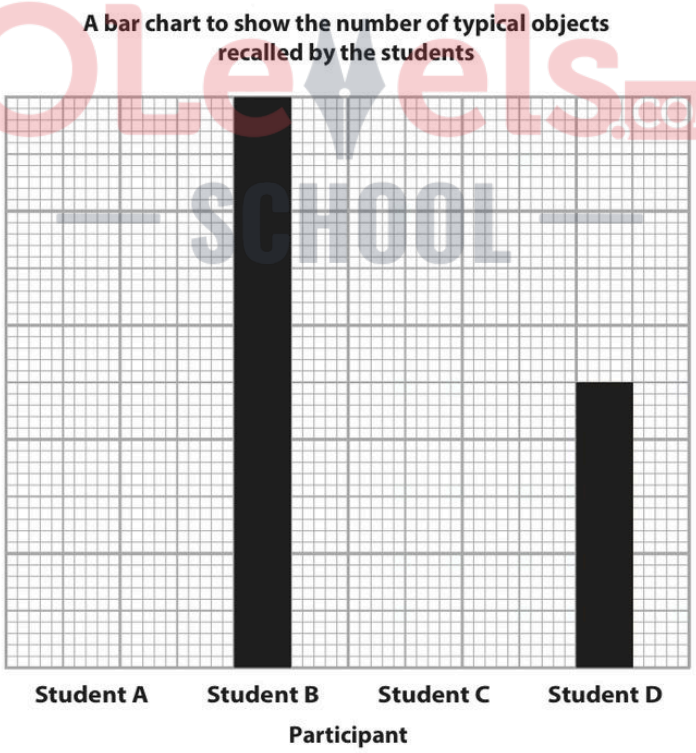
(b) Calculate how many students recalled **less than five unusual objects** as a fraction of all students.

(1)

Fraction .....

(c) Complete the bar chart using the data in **Table 1**, including the label and scale for the y-axis.

(3)



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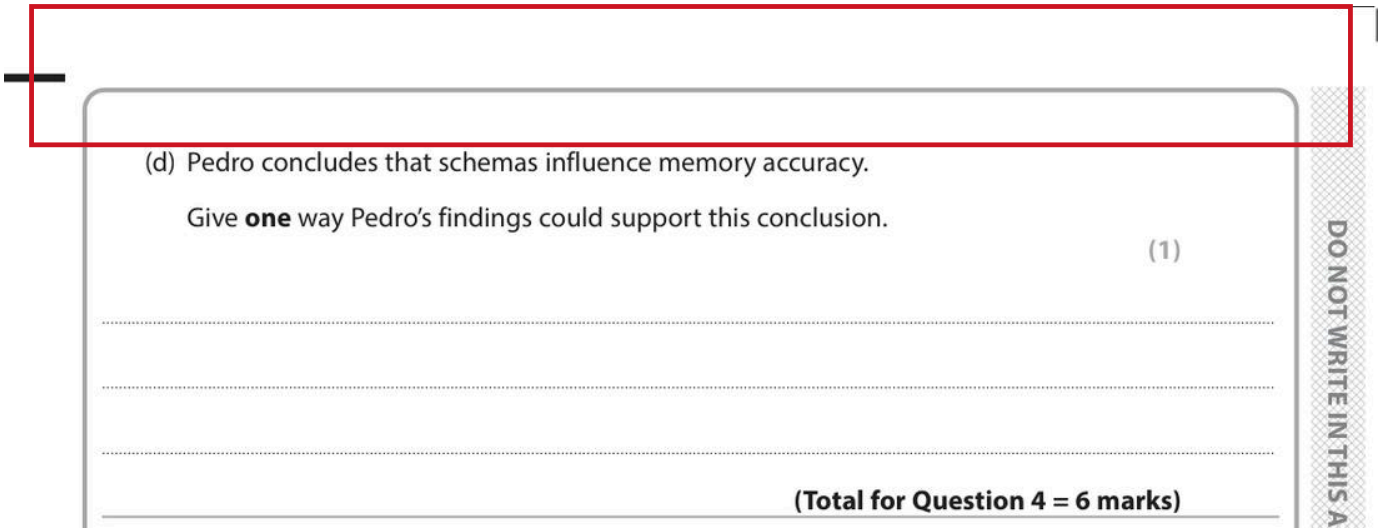
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Turn over

8. Question 4 · part c · 3 marks

2024 JUN · Paper 02

Further practice · Attempt part c only.



(d) Pedro concludes that schemas influence memory accuracy.

Give **one** way Pedro's findings could support this conclusion.

(1)

(Total for Question 4 = 6 marks)

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Working and answer

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9. Question 7 · part b · 2 marks

2020 NOV · Paper 01

Further practice · Attempt part b only.

7 Lara wanted to investigate different forms of revision with her Geography classes.

Class 1 and class 2 were given a list of 12 capital cities to learn for a Geography test.

Class 1 were given grouped capital cities with the same first letter.

Class 2 were given ungrouped capital cities.

Figure 2 shows the lists given to class 1 and class 2.



Figure 2

Class 1 scored higher on average for the Geography test.

(a) Explain **one** reason for the higher performance of class 1 compared to class 2. You should refer to the Multi-store Model of Memory in your answer.

(2)

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9. Question 7 · part b · 2 marks

2020 NOV · Paper 01

Further practice · Attempt part b only.

The performance of the students in class 1 is shown in **Figure 3**.

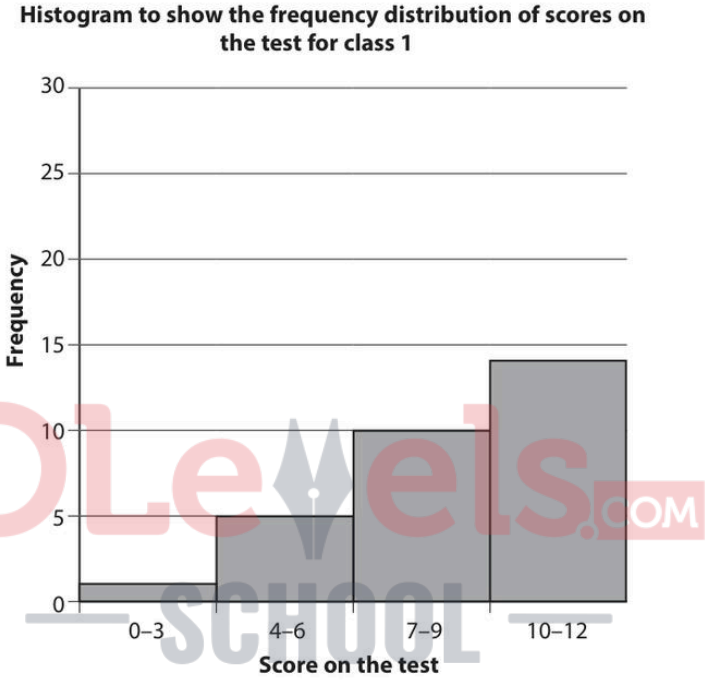


Figure 3

(b) Explain **one** conclusion that can be made regarding the type of distribution shown in **Figure 3**.

(2)

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9. Question 7 · part b · 2 marks

2020 NOV · Paper 01

Further practice · Attempt part b only.

(c) Explain **one** improvement that Lara could make to her investigation.

(2)

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(Total for Question 7 = 6 marks)

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Working and answer